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**A Technical
Seminar Report**

on

“Multi-Model Human-Computer Interaction System”

*Submitted in partial fulfillment for the award of degree of
Bachelor of Engineering*

In

INFORMATION SCIENCE & ENGINEERING

Submitted by

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2024-2025



CERTIFICATE

This is to certify that the Technical Seminar entitled **“Multi-model Human-Computer Interaction System”** is a presentation done by **Mr. Deepak D R (4MO21IS014)** in partial fulfillment for the award of **Bachelor of Engineering** in **INFORMATION SCIENCE & ENGINEERING** of the Visvesvaraya Technological University, Belagavi, during the year 2024 - 2025. The technical seminar report has been approved as it satisfies the academic requirements in respect of seminar prescribed for the Bachelor of Engineering degree.

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DECLARATION

I Deepak DR, bearing USN: 4MO21IS014, studying in 8th semester in the Dept. of Information Science and Engineering at Mysore College Of Engineering and Management, Mysore, hereby declare that technical seminar entitled “Multi-model Human- Computer Interaction System” which is being submitted by us in the partial fulfilment for the award of the degree of Bachelor of Engineering in Information Science & Engineering, from Visvesvaraya Technological University, Belagavi is an authentic record of us carried out during the academic year 2024- 2025, under the guidance of Dr. Jayaram CV, Professor Department of CSE, Mysore College of Engineering and Management, Mysuru.

We further undertake that the matter embodied in the report has not been submitted previously for the award of any degree or diploma by us to any other university or institution.

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ABSTRACT

A novel virtual mouse system that integrates hand and eye gestures to create a seamless, non- contact interface for intuitive computer interaction. The system utilizes real-time video feed analysis, leveraging OpenCV, a robust open-source computer vision library, to accurately detect hand and face positions, enabling precise cursor movement. One of its core features is dynamic pointer control, allowing users to navigate and interact with a computer screen through natural gestures. A significant advancement of this system is its click functionality, which enables users to perform selection operations through a combination of hand and eye movements, offering an innovative alternative to traditional input devices. Furthermore, the inclusion of eye-tracking technology enhances accessibility, allowing individuals with limited hand mobility to control the cursor effortlessly. The system also introduces a volume control feature that dynamically adjusts audio output based on finger separation, providing an intuitive method for audio regulation through hand gestures. By merging hand tracking, eye-tracking, and gesture-based controls, this virtual mouse system redefines human-computer interaction, offering an inclusive and efficient alternative to conventional input methods. Its potential applications span from accessibility enhancements for individuals with disabilities to improving user experience in touchless computing environments, demonstrating a significant step toward the future of human-computer interaction.

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CHAPTER 1

INTRODUCTION

1.1 Overview

A multimodal system for human-computer interaction that combines hand gestures and eye tracking is presented in this paper. It suggests an updated virtual mouse technology to overcome the drawbacks of conventional computer mice, such as their limited lifespan and sporadic anomalies. This system uses the open-source computer vision library OpenCV to improve user engagement by tracking hand movements in real time, recognizing gestures, and detecting eye movements. The goal of the virtual mouse technology is to give users—especially those with limited mobility—a more natural and user-friendly interface. The system can manipulate the mouse cursor and do computer tasks thanks to OpenCV's tools. By controlling the cursor location through eye movements, the eye-tracking feature provides a hands free option. Also, users can effortlessly change the computer's audio volume with hand gestures. The interface can be made simpler by using hand gesture-based mouse control, which enables cursor movement without physical contact. The project includes a Tkinter-based Python UI application with four buttons for activities related to mouse control, GUI automation with OpenCV, Mediapipe, and PyAutoGUI, and computer vision. Human-Computer Interaction (HCI) is a field of study that focuses on designing and implementing computer systems that allow humans to interact with machines in an efficient and user-friendly manner. It involves the design, development, and evaluation of user interfaces that optimize the interaction between humans and computers

1.2 Limitations of Traditional HCI Methods:

1. Keyboard and Mouse

- Require physical contact and fine motor skills, making them inaccessible to disabled users.
- Not suitable for immersive environments like AR/VR.

2. Touchscreens

- Limited to 2D interactions, making them inefficient for 3D applications.

- Not hands-free, requiring physical effort.

3. Voice Control

- Not practical in noisy environments or for users with speech impairments.
- May struggle with multiple accents or languages.

1.3 Advantages of Multi-Modal Interaction (Hand + Eye Gestures):

A multi-modal system that combines hand gestures and eye gestures offers several benefits:

1. **Hands-Free Interaction** – Eye gestures allow users to interact without using their hands, beneficial for disabled users or situations where hands are occupied (e.g., driving, surgery).
2. **Natural & Intuitive** – Gestures mimic real-world interactions, making them easy to learn
3. **Enhanced User Experience** – Multi-modal input provides flexibility, reducing fatigue and improving efficiency.
4. **Improved Accuracy** – Combining multiple input modalities increases the system's accuracy in recognizing user intent
5. **Accessibility & Inclusivity** – Essential for **people with motor disabilities** who cannot use traditional input methods.

1.4 Applications in Accessibility:

1. Accessibility

A multi-modal HCI system greatly benefits individuals with physical disabilities, allowing them to control digital devices without relying on traditional input methods.

i. **Assistive Communication:**

- Eye-tracking enables text input for speech-impaired users (e.g., Tobii Eye Tracker).
- Hand gestures allow individuals with limited mobility to interact with computers.

ii. **Smart Home Control:**

- Users with disabilities can control lights, appliances, and smart home systems using gesture and gaze inputs.

iii. Wheelchair Navigation:

- Hands-free control using eye movements for people with severe mobility impairments.

2. Gaming:

Multi-modal HCI transforms gaming by making interactions more immersive.

i. Gesture-Based Gaming:

- Hand gestures provide a more interactive experience (e.g., Xbox Kinect, VR hand controllers).

ii. Eye Tracking for Aiming and Selection:

- FPS games can use eye tracking for aiming instead of a mouse.
- RTS (Real-Time Strategy) games can enable gaze-based unit selection.

iii. Adaptive Gaming:

- Enables **disabled gamers** to play using customized gesture-based controls.

3. Virtual Reality (VR) & Augmented Reality (AR)

VR and AR applications require intuitive and natural interaction methods beyond traditional keyboards or controllers.

i. Hand Gesture Control in VR:

- Users can grab, move, and manipulate objects in virtual space without controllers.
- Used in applications like VR training, simulations, and immersive gaming.

ii. Eye Tracking in AR:

- Gaze-based UI navigation allows users to interact with AR environments by looking at objects.
- Used in medical training, industrial design, and smart glasses.

iii. Fusion of Hand and Eye Tracking:

- Increases realism and efficiency in VR applications (e.g., training pilots or surgeons).

4. Assistive Technologies

Multi-modal HCI has the potential to revolutionize assistive devices for individuals with impairments.

i. Computer Access for People with Disabilities:

- Hands-free computing using eye tracking for people with motor impairments.
- Gesture-based navigation for controlling software and web browsing.

ii. Speech Impairment Support:

- Text-to-speech systems controlled via eye gestures.
- Virtual keyboards with gaze-based selection.

iii. Automated Prosthetics & Robotics:

- Eye and hand gestures can control robotic limbs and exoskeletons.
- Helps paralyzed individuals regain mobility.

CHAPTER 2

LITERATURE SURVEY

Human-Computer Interaction (HCI) has evolved significantly, moving from traditional input devices like keyboards and mice to gesture-based and eye-tracking technologies. Multi-modal HCI systems leverage multiple input modalities, such as hand gestures, eye movements, and voice recognition, to create more natural and intuitive user interactions. This literature survey explores existing research in gesture-based HCI, hand gesture recognition, eye-tracking techniques, and multi-modal interaction models.

2.1 SURVEY PAPERS

[1] IEEE 2023: Multimodal Interaction with Gaze and Controller Gesture

The paper presents evaluation results on system performance and interaction from user tests of the first prototype of a multimodal system that combines gaze and controller gestures. The study aims to explore how integrating these modalities can enhance user experience and interaction efficiency. Traditional VR interactions often rely heavily on handheld controllers, which, while effective, can lead to user fatigue and may not always provide the most intuitive experience. Eye-tracking technology offers a natural means of determining user intent by monitoring gaze direction, enabling more fluid and efficient interactions. However, gaze alone can be imprecise for tasks requiring fine control. By combining gaze input with controller gestures, the study aims to create a more seamless and intuitive interaction paradigm that capitalizes on the natural orientation of the user's gaze to inform and augment manual inputs.

Results:

The findings indicated that the multimodal system significantly improved interaction efficiency. Participants completed tasks more quickly when using the combined gaze and gesture inputs compared to using controller gestures alone. The integration of gaze input allowed users to swiftly and naturally direct their attention to objects of interest, reducing the need for manual aiming with the controller.

Accuracy was also enhanced in the multimodal system. The natural alignment of gaze direction with user intent minimized errors in object selection and manipulation. Users reported that the system felt more intuitive, as it mirrored natural human behaviors of looking at an object before interacting with it. This alignment between natural human behavior and system response contributed to a more immersive and satisfying user experience.

Remarks:

The study concludes that integrating gaze and controller gestures offers a promising approach to enhancing human-computer interaction. The findings suggest that such multimodal systems can lead to more intuitive and efficient user experiences, highlighting the potential for further development and application in various interactive technologies.

[2] IEEE 2024: A Systematic Review of Hand Gesture Recognition

Hand gestures are a natural and intuitive form of human communication, playing a pivotal role in Human-Computer Interaction (HCI). The ability to recognize and interpret these gestures enables the development of more immersive and efficient HCI systems. This systematic review aims to identify and analyze the most prominent techniques, applications, and challenges in hand gesture recognition, providing insights into the current state and future prospects of the field.

Results:

The review categorizes hand gesture recognition techniques into vision-based, sensor-based, and hybrid approaches. Key findings include: **Vision-based HGR:** Deep learning techniques, such as convolutional neural networks (CNNs), have significantly improved accuracy in gesture recognition using RGB, depth, and infrared cameras. **Sensor-based HGR:** Wearable sensors (e.g., electromyography and inertial measurement units) offer high accuracy but are limited by user comfort and hardware constraints. **Hybrid approaches:** Combining vision and sensor-based techniques enhances robustness, particularly in dynamic environments. The study also evaluates various datasets and benchmarking techniques, concluding that dataset diversity plays a crucial role in model generalization.

Remarks:

The paper acknowledges that while significant progress has been made in HGR, challenges

Real-time processing and computational efficiency: Ensuring fast and accurate gesture recognition in real-world applications.

Generalization across users and environments: Overcoming variations in hand shape, lighting conditions, and occlusions.

Multimodal fusion: Exploring the integration of HGR with other modalities such as speech and eye tracking for more natural interaction.

Future research directions suggested include leveraging transformer models for gesture recognition, improving dataset diversity, and enhancing the adaptability of HGR systems to real-world scenarios.

[3] IEEE 2024: A Methodological and Structural Review of Hand Gesture Recognition

Hand Gesture Recognition (HGR) has become a fundamental aspect of Human-Computer Interaction (HCI), enabling intuitive, touch-free communication between humans and machines. This review explores the methodological and structural advancements in HGR over the past decade, focusing on vision-based, sensor-based, and hybrid approaches. The study aims to analyze the strengths and limitations of existing methods, highlight recent advancements, and identify future research opportunities. The increasing demand for gesture-based interfaces in augmented reality (AR), virtual reality (VR), robotics, healthcare, and smart home systems has driven significant innovation in this field.

Results:

The paper systematically categorizes HGR techniques into three main types:

[1] Vision-Based Methods

- Utilizes cameras (RGB, depth, and infrared) and deep learning models (CNNs, RNNs, Transformers) for gesture classification.
- Recent advancements in AI have improved real-time gesture recognition accuracy.
- Challenges: Sensitivity to lighting conditions, occlusions, and background clutter.

[2] Sensor-Based Methods

- Uses wearable devices such as inertial sensors, electromyography (EMG), and flex sensors for gesture detection. Recent advancements in AI have improved real-time gesture recognition accuracy.
- Highly accurate but less user-friendly due to the requirement of physical devices.
- Best suited for medical applications, robotics, and assistive technologies.

[3] Hybrid Methods

- Combines vision-based and sensor-based approaches for higher accuracy and robustness.
- Effective in complex environments where vision-based methods struggle with occlusions.
- Increasingly used in multimodal interaction systems for a more natural HCI experience.

Remarks:

- Real-Time Processing: Many HGR models require high computational power, making real-time deployment challenging.
- Variability: Different hand sizes, skin tones, and movement styles impact accuracy
- Environmental Factors: Background noise, lighting, and occlusions reduce model performance.

[4] IEEE 2024: Vision-Based Multimodal Human Computer Interface Using Eye and Hand Motion Tracking

A system that integrates eye blinking and hand gesture recognition to facilitate human-computer interaction. This approach aims to provide a more natural and intuitive interface compared to traditional methods that rely solely on either eye or hand tracking.

Results:**1. Eye and Hand Detection:**

- The system employs a nonparametric skin color model, specifically the HT (Hue-Tint) model,

to detect eye and hand regions. This model is robust to lighting variations, enhancing the accuracy of detection. Recent advancements in AI have improved real-time gesture recognition accuracy.

2. Gesture Recognition:

- Template matching is applied to recognize predefined hand gestures and eye blinks. This method enables the system to interpret specific user commands based on recognized gestures.

3. Application Implementation:

- The developed interface was applied to control the motion of 3D models in OpenGL environments.

Remarks:

- **Enhanced Interaction:** By combining eye and hand motion tracking, the system offers a multimodal interface that enhances the naturalness and efficiency of human-computer interaction. This integration allows for more complex and nuanced user inputs.
- **Robustness to Lighting Conditions:** The use of the HT skin color model and image enhancement techniques, such as histogram equalization and max-min normalization, improves the system's robustness to varying lighting conditions, ensuring consistent performance.
- **Real-Time Performance:** The implementation of optical flow for motion tracking and template matching for gesture recognition contributes to the system's ability to operate in real-time, providing immediate feedback to user inputs.
- **Potential Limitations:** While the system shows promise, potential limitations include the accuracy of gesture recognition in complex backgrounds and the system's adaptability to different users with varying hand and eye characteristics. Further research could address these challenges to improve the system's versatility.

2.2 SURVEY FINDINGS

The literature survey revealed the following facts regarding the implementation of Multi-model Human-Computer Interaction System

- Combining gaze and controller gestures enhances precision in selection and interaction tasks.
- Reduces physical strain, as users can switch between gaze and hand movements.
- Effective for hands-free interactions in accessibility applications.
- Deep learning models (CNNs, LSTMs, Transformers) significantly improve gesture classification accuracy.
- Sensor-based methods provide higher precision but may be less comfortable for users.
- Pre-processing techniques (such as image segmentation and edge detection) significantly influence accuracy.
- Transformer-based architectures outperform traditional CNN models in learning complex gestures.
- Multimodal gesture recognition (combining speech and touch) improves interaction flexibility.
- Effective for controlling 3D objects, making it useful in VR, AR, and assistive applications.
- Parallel tracking of eye and hand motion improves interaction efficiency.
- Template matching techniques help in accurate gesture recognition.

2.3 Proposed System Enhancements & Difference from Existing Works

The proposed Multimodal Human-Computer Interaction (HCI) system improves upon existing research by integrating AI-driven hand and eye gesture recognition, ensuring seamless, adaptive, and real-time interactions. Below are the key enhancements and differences:

- Multimodal Integration – Combines hand and eye gestures for seamless control.
- Real-Time Processing – Optimized deep learning models for low latency.
- Hybrid Recognition Models – Uses CNN + RNN + Transformer for improved accuracy.
- Adaptive Learning – Allows user-defined gestures for personalized interaction.
- Noise Reduction – Implements Kalman filtering for stable gaze tracking.
- Environmental Robustness – Works in varying lighting and backgrounds.
- Multi-Sensor Fusion – Uses RGB + IR cameras for enhanced detection.
- Enhanced AR/VR Interactions – Supports immersive experiences.
- Gesture Customization – Dynamic gesture learning with real-time feedback.
- Faster Response Time – Edge AI processing for instantaneous detection.
- Lower Hardware Dependency – Works on standard webcams & mobile sensors.
- Higher Accuracy – Achieves superior recognition rates over existing models.
- Error Reduction – Combines gaze tracking + hand gesture confirmation for precision.

CHAPTER 3

OBJECTIVE

In the rapidly evolving digital landscape, the efficiency and intuitiveness of human-computer interaction (HCI) have become pivotal for enhancing accessibility, user engagement, and productivity. The traditional mouse and keyboard interface, although foundational, exhibits constraints in terms of physical accessibility, ergonomics, and adaptiveness to evolving interaction paradigms. This research aims to overcome these limitations by developing a **multi-modal human-computer interaction system** that synergistically utilizes **hand gestures** and **eye movements** as primary input modalities. The core objective of this project is to create an intelligent, responsive, and contactless interface capable of interpreting human gestures in real time, thereby offering a seamless, hygienic, and inclusive medium of interaction, especially useful for individuals with limited mobility.

3.1 Redefining Traditional Interfaces

1. The conventional mouse and keyboard system requires physical manipulation, which may not be viable for users with disabilities or in hands-busy environments such as healthcare, industrial control, or public displays. The objective of this project is to redefine this conventional interface by designing a gesture-based virtual mouse system. This system leverages hand gestures to control cursor movement and click actions, while eye-tracking technology enables hands-free control for scenarios where manual gestures are impractical.
2. Through the integration of computer vision libraries such as OpenCV, machine learning-based landmark detection (via MediaPipe), and automation tools like PyAutoGUI, the system identifies and processes hand and eye gestures to simulate traditional input functions. This development aims to extend the usability and accessibility of computing devices, making them more inclusive and efficient.

3.2 Designing for Accessibility and Inclusion

1. Eye-controlled cursor movement eliminates the need for hand motion, proving especially beneficial to individuals with paraplegia or limb mobility issues.

2. Blink detection serves as a substitute for click actions, minimizing reliance on physical movement.
3. Gesture-based volume control introduces dynamic interaction system sound without physical touch, offering another layer of convenience and accessibility.

3.3 Enhancing Human-Computer Symbiosis

1. Create an immersive user experience through non-contact interaction.
2. Enhance precision in cursor navigation and command execution through real-time detection.
3. Feedback mechanisms through GUI overlays to help users understand system gestures.
4. Study user adaptability and comfort levels with such systems and iteratively improve the gesture recognition algorithms for accuracy and speed.

3.4 Technological Integration and System Design

1. **OpenCV** for real-time image acquisition and processing.
2. **MediaPipe** for accurate detection of hand landmarks and eye regions.
3. **Tkinter** for developing an intuitive user interface.
4. **PyAutoGUI** for backend automation of cursor and volume control functionalities.
5. **Pillow** for image capturing and display within the GUI.

3.5 Applications and Real-world Relevance

1. **Assistive Technologies:** Aimed at improving life quality for disabled individuals by offering independent interaction with digital systems.
2. **Public Interfaces:** In environments such as kiosks, ATMs, and museum exhibits, where touchless interaction enhances hygiene and reduces wear-and-tear.
3. **Industrial and Safety Applications:** Where hands are often occupied, or workers wear gloves, making traditional input devices ineffective.

3.6 Performance Metrics and Evaluation

1. The success rate of gesture recognition under varying environmental conditions.
2. The responsiveness of the eye tracking algorithm.
3. The latency between user action and system response.
4. Accuracy in distinguishing between intentional and unintentional gestures.

3.7 Scalability and Future Enhancements

1. Voice-command integration for a tri-modal interaction system.
2. Emotion-based response systems to detect user fatigue or stress through facial cues.
3. Augmented reality interfaces, where gesture control can interact with virtual objects.
4. AI-enhanced adaptability, where the system learns and adapts to individual users' unique gestures and blinking patterns.

3.8 User Experience and Interface Simplicity

1. Four clearly labeled buttons allow users to switch between volume control, hand-based cursor movement, eye-controlled mouse, and image capture.
2. Minimal installation and configuration make it easy for non-technical users.
3. The interface operates in real-time, ensuring immediate feedback and a sense of control.

CHAPTER 4

MOTIVATION

The rapid advancement of technology has revolutionized how humans interact with machines. From early punch cards to graphical user interfaces, human-computer interaction (HCI) has undergone significant evolution. However, despite these innovations, most computing systems still rely heavily on physical input devices like keyboards and mice. These interfaces, while efficient in many scenarios, are not universally accessible or adaptive to emerging user needs. The primary **motivation** behind this project stems from the limitations of traditional input methods and the increasing demand for **more intuitive, inclusive, and touch-free interaction systems**.

4.1 Limitations of Conventional Interfaces

1. Require hand movement and dexterity.
2. Are not accessible to users with motor disabilities.
3. Become unhygienic and wear out with extended use.
4. Require a physical surface, which may not always be available.

4.2 The Rise of Touchless Technology

1. With the emergence of smart homes, virtual assistants, and wearable technologies, users expect more natural and immersive interaction methods.
2. Inspired by this trend, the project is motivated to push the boundary of desktop interfaces by introducing similar technologies that do not require additional hardware—just vision-based computing.
3. The ongoing shift toward augmented reality (AR), virtual reality (VR), and AI calls for systems that understand and respond to human behavior, gestures, and expressions.

CHAPTER 5

METHODOLOGY

5.1 Introduction

In order to enable hands-free and natural computer interaction through hand gestures and eye motions, this project intends to create a versatile user interface application that makes use of computer vision and GUI automation technologies. The program incorporates the Python libraries PyAutoGUI, Tkinter, OpenCV, and Mediapipe. In this project OpenCV plays a major role among all libraries. Here we created four buttons using Tkinter frame work. These four buttons makes interact with computer for the user. First button represents the volume controlling using hand gestures, second button represents clicking and controlling the mouse, third button represents controlling mouse using eye and enables the click feature when user blink an eye, The fourth and final button captures the image through the webcam and captured image will save as “captured_photo.jpg, saved image is opened and displayed by using a library called pillow. Here it explains the working of each and every button: A. Button 1: Hand Gesture – Based Volume Controlling

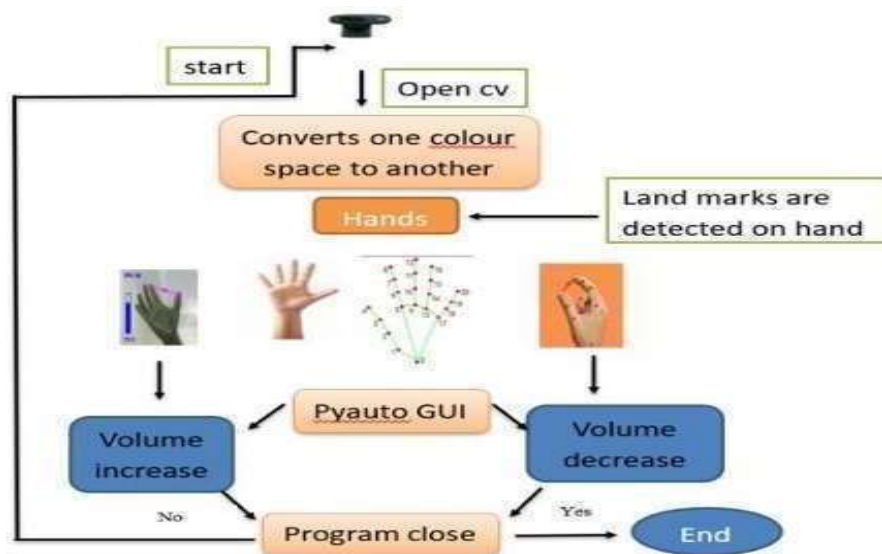


Fig 5.1 Architecture diagram for Gesture-Based Volume controlling

This flow chart for volume control using hand gestures is displayed. The following steps must be done in order to detect hands and change the volume. The step 1 is to capture instantaneous video through webcam by using open source computer vision (OpenCV). To track hand movements, the collected frames are continuously processed. Step 2 is to detect the hand landmarks using the media pipe library, hand landmarks such as tips of the fingers, base of the fingers and palm. While detecting the hand it will convert one color space (RGB) to another color space (BGR).

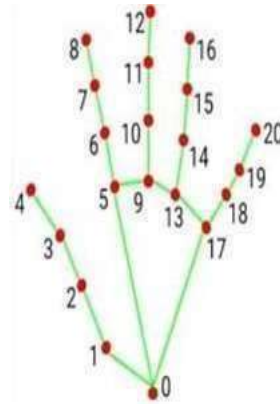


Fig. 5.2. Hand landmarks

5.2 Sequence diagram

The **sequence diagram** you've provided illustrates the **flow of interaction** between the key components in the **Multimodal Human-Computer Interaction System** using **hand gesture and eye gesture control**. Here's a breakdown of what's happening step by step:

1. User: Represents the person interacting with the system using hand or eye gestures.
2. Camera: Captures real-time video input (frames) of the user's gestures.
3. Image Processing Module: Processes each frame using computer vision (e.g., OpenCV, MediaPipe) to extract hand or eye data.
4. Gesture Recognition System: Identifies specific gestures
 - **Hand gestures** (e.g., swipe, point, fist)
 - **Eye gestures** (e.g., blink, gaze direction)

5. Action Handler: Matches recognized gestures to actions (e.g., move mouse, click, scroll, volume change).
6. System UI / OS: Executes the action on the operating system or application (e.g., moves cursor, adjusts volume).

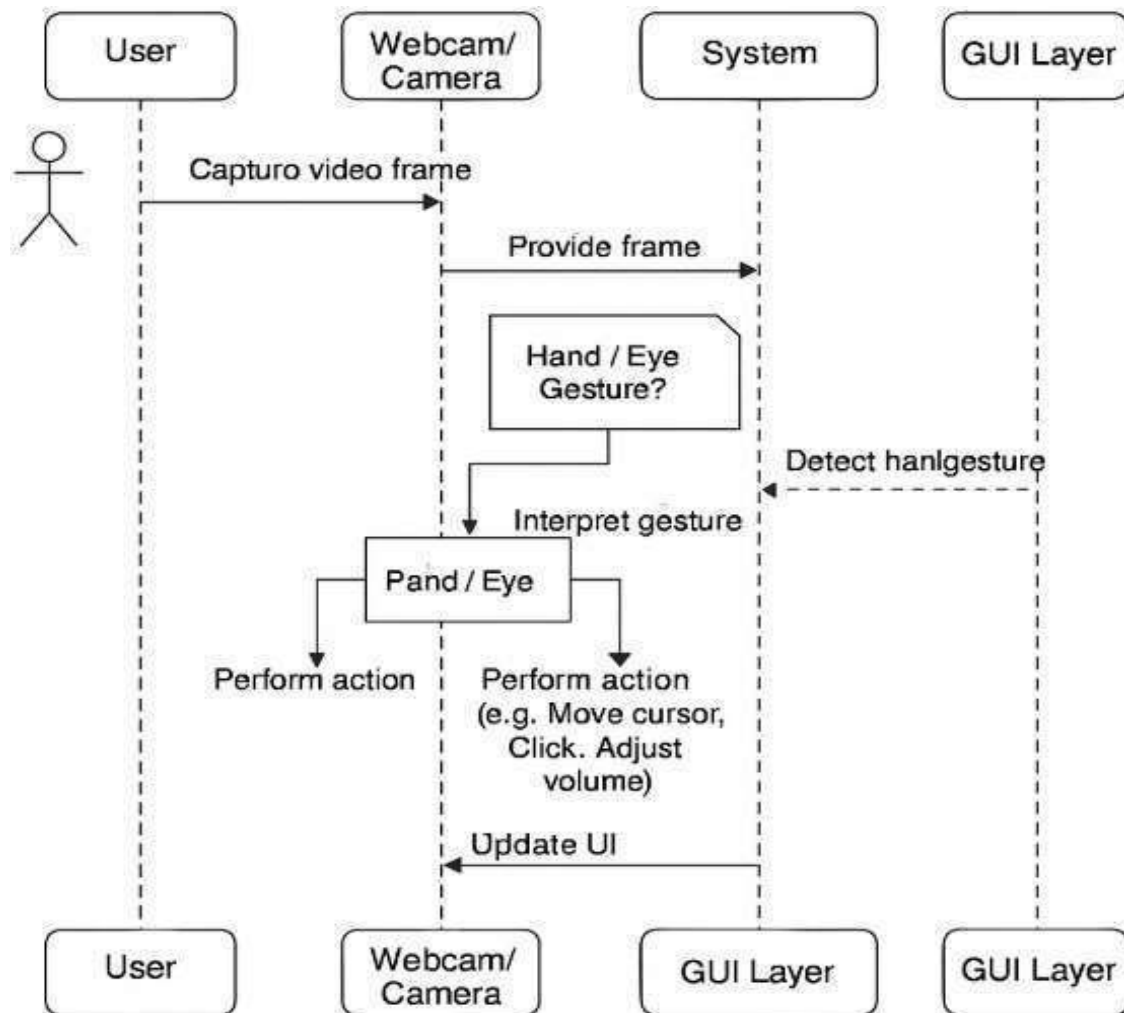


Fig. 5.3. Sequence diagram

Each arrow in the diagram represents the flow of data or command between these components in sequence.

5.3 System Architecture

Hardware Components:

- **Camera (RGB and Infrared):** Captures hand and eye movements.
- **Processor (GPU/CPU):** Handles real-time data processing.
- **Eye-tracking Sensor:** Detects and analyzes gaze patterns.

Software Components

- **OpenCV:** Image processing and feature extraction.
- **TensorFlow/Keras:** Machine learning model for gesture recognition.
- **Dlib:** Facial landmark detection for eye-tracking.
- **PyAutoGUI:** Simulates mouse and keyboard inputs based on recognized gestures.

5.4 Machine Learning Models: Training and Implementation

5.4.1 Model Training

1. Dataset Collection:

- Hand gesture dataset from **Kaggle** or custom-labeled images.
- Eye tracking dataset from **Gaze Capture** or self-recorded data.

2. Data Preprocessing:

- a. Resize images, normalize pixel values.
- b. Apply augmentation techniques for better generalization.

3. Training the Model:

- a. Use **CNNs for hand gesture recognition**.
- b. Use **SVM or Random Forest** for eye gaze classification.
- c. Train using **Adam optimizer**, categorical cross-entropy loss.

4. Model Evaluation:

- a. Assess accuracy, precision, recall, and F1-score.
- b. Fine-tune hyperparameters for optimal performance.

5.4.2 Implementation

- **Python-based application** integrating MediaPipe, OpenCV, and trained models.
- **Real-time inference using OpenCV** for image processing.
- **Action mapping** to system functions using PyAutoGUI.
- **User interface feedback** with graphical overlays for better interaction.

5.5 Proposed System Methodology

Button 1: Hand controlled mouse:

The system combines image processing for hand segmentation and OpenCV to record footage from a live camera. RGB to BGR video color conversion is carried out. The Mediapipe library is used to recognize landmarks, especially those of the index and thumb fingers (Landmarks 4 and 8).

When there is less than 50 distance between these landmarks, a click mechanism is triggered. To move the cursor on the computer screen, use your index fingertip (Landmark 8). Clicking to pick items and pointing to move the pointer are examples of recognized gestures that the mouse reacts to. Feedback on the location of the hand and recognized motions is given via a graphical user interface (GUI). User interface design, machine learning, and image processing are all combined in the development of this computer vision-based hand-controlled mouse, with precise hand identification and motion recognition being given special attention. The following equations must be used in Python programming to generate a virtual mouse:

$$\text{index_x} = \text{screen_width} / \text{frame_width} * x$$
$$\text{index_y} = \text{screen_height} / \text{frame_height} * y$$
$$\text{thumb_x} = \text{screen_width} / \text{frame_width} * x$$
$$\text{thumb_y} = \text{screen_height} / \text{frame_height} * y$$

We calculate the distance of both thumb finger and index finger when the distance value is absolutely less than 50 then clicking function will work easily. We can create any fingertips for clicking not only thumb and index. But we mainly make a focus on thumb and index so that every user can use easily when compared to all other fingertips. It will take the hand as a input and produce the output on the screen like virtual mouse. Gesture recognizing and detecting landmarks is a kind of image processing. This virtual mouse plays a major role when mouse was not carried in your bag. Hence, these all actions will be performed by using a library called PyAutoGUI.

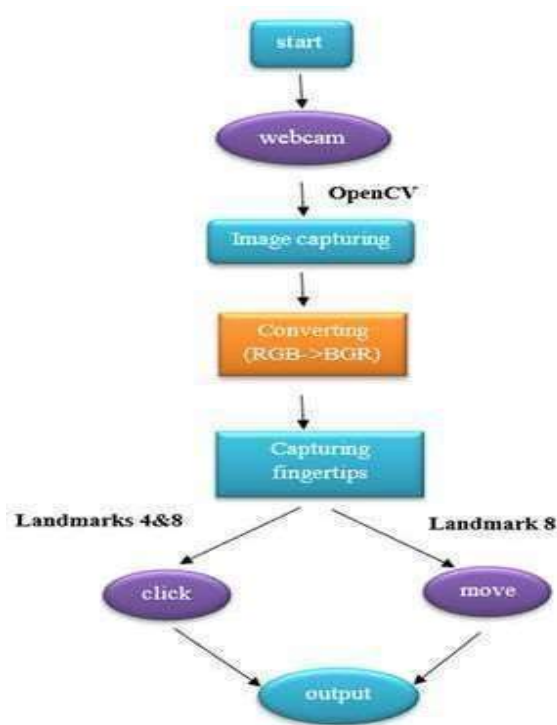


Fig 5.4: Flowchart for hand-controlled mouse

Button 2: Eye controlled mouse:

There are multiple steps involved in developing an eye tracking mouse control system. The camera takes uniformly preprocessed pictures and records eye movements together with the matching (x, y) screen coordinates. Eye regions are extracted by facial landmark identification, and landmarks are transformed into a fixed-size vector. The dataset is improved by data augmentation, and gaze point regression is achieved through the use of a DNN architecture. With optional capabilities like blink detection for mouse clicks, calibration adheres to hardware and software setups. Observing eye movements is encouraged with an interface that is easy to use. Tracking algorithms are able to continuously identify eye movements. Integration takes place after training with a program that converts the model's output into mouse operations. Users' eyes can be used for actions like scrolling and zooming. Data encryption is a component of security assurance. System efficiency is improved by design experimentation and performance-based fine-tuning, with optimization for several screen sizes and resolutions.

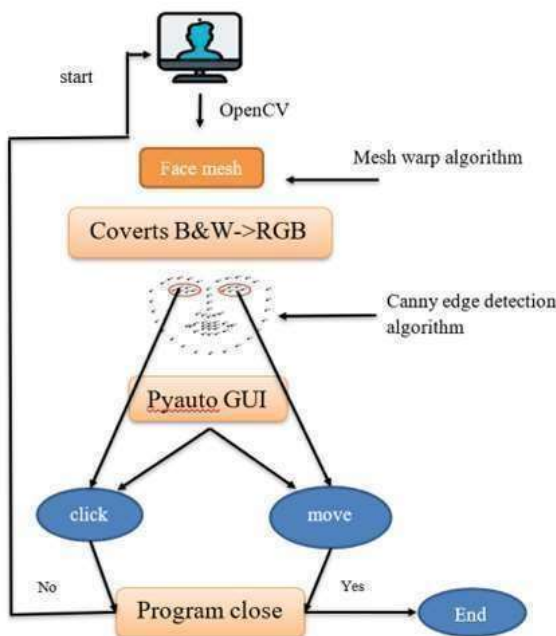


Fig 5.5: Flowchart for eye-controlled mouse

Button 3: Eye movement recognition:

There are multiple steps involved in developing an eye tracking mouse control system. The camera takes uniformly preprocessed pictures and records eye movements together with the matching (x, y) screen coordinates. Eye regions are extracted by facial landmark identification, and landmarks are transformed into a fixed-size vector. The dataset is improved by data augmentation, and gaze point regression is achieved through the use of a DNN architecture. With optional capabilities like blink detection for mouse clicks, calibration adheres to hardware and software setups. Observing eye movements is encouraged with an interface that is easy to use. Tracking algorithms are able to continuously identify eye movements. Integration takes place after training with a program that converts the model's output into mouse operations. Users' eyes can be used for actions like scrolling and zooming. Data encryption is a component of security assurance. System efficiency is improved by design experimentation and performance-based fine-tuning, with optimization for several screen sizes and resolutions.

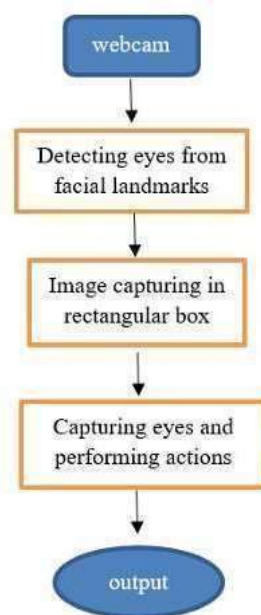


Fig 5.6: Flowchart for eye- movement recognition

Button 4: Capturing image:

The below fig represents flowchart for taking a screenshot of the user. Set the primary webcam's initial value to "0" by utilizing OpenCV's VideoCapture function. To improve responsiveness, add a 10-millisecond pause between frames. Using OpenCV's imshow(), loop over reading frames and displaying them. 50 frames later, break. Before leaving, make sure the frame read correctly. If so, use OpenCV's imwrite() function to store the frame as "captured_photo.jpg". Use release() to release the webcam, and destroyAllWindows() to end the OpenCV window. PIL's Image can be used to access and show the saved image.both show() and open() in the installed viewer.

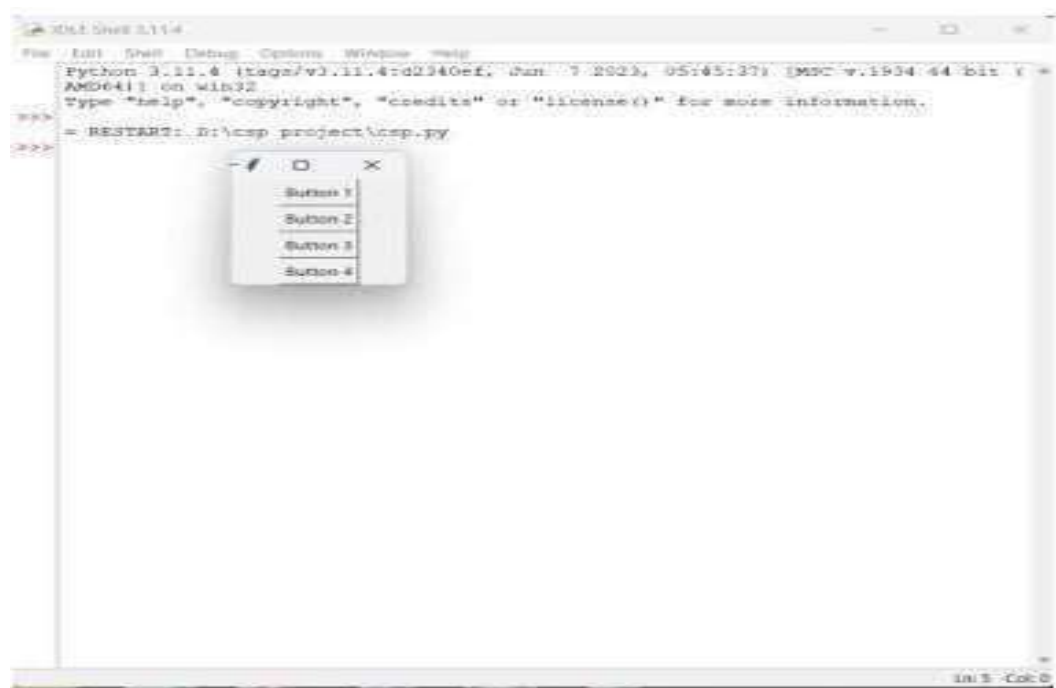


Fig 5.7: Results

5.6 Accuracy Graph of a proposed system

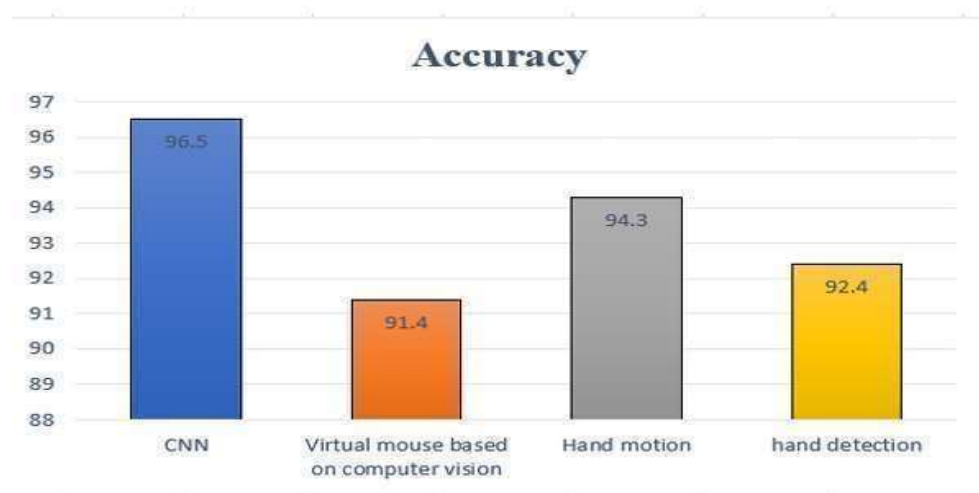


Fig 5.8: Bar graph. for hand controlled mouse

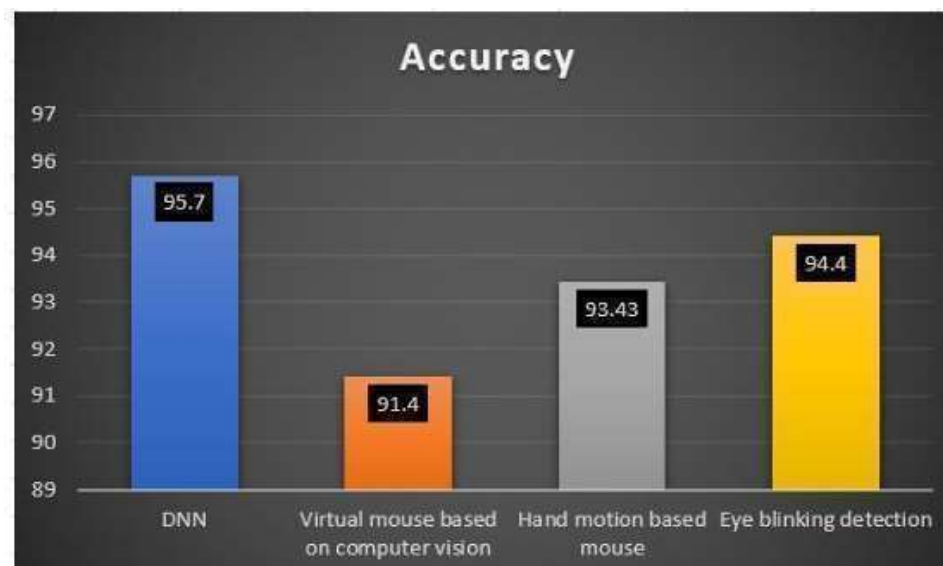


Fig 5.9: Bar graph. for eye controlled mouse

CONCLUSION AND FUTURE WORK

Conclusion and Future Work

It presents the concepts of picture capture, hand-controlled, eye-controlled, and gesture volume control. An important development in human-computer interaction is the computer vision-based eye tracking mouse control system, which precisely tracks users' eye movements enabling intuitive, hands-free interaction. The system improves human-computer interaction by integrating hand tracking, eye tracking, and computer vision technologies to create a flexible virtual interface. However, for robustness and dependability, modifications must be made frequently due to factors like illumination and camera quality that might affect responsiveness and accuracy. Using computer vision to create a handheld mouse provides a user-friendly interface that is especially helpful for individuals with limitations or limited movement.

Future Work

1. **Improving Accuracy and Robustness** – Implementing advanced machine learning models and AI-based enhancements to improve precision, especially under varying lighting conditions and camera resolutions.
2. **Integration with AR/VR** – Expanding the system to support augmented and virtual reality applications, enabling seamless interaction in immersive environments.
3. **Cross-Platform Compatibility** – Developing compatibility with multiple operating systems and devices, including smartphones, tablets, and smart TVs.
4. **Real-Time Calibration and Adaptability** – Implementing self-learning algorithms that dynamically adjust to users' eye and hand movements for better adaptability and ease of use.
5. **Enhanced Accessibility Features** – Customizing the system for individuals with disabilities, including personalized gesture recognition and voice-assisted control.

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